

EXAM 7 – SPRING 2015

1. (2.25 points)

Given the following information:

<u>Incremental Paid Claims (\$000)</u>				
Accident <u>Year</u>	0-12 <u>Months</u>	12-24 <u>Months</u>	24-36 <u>Months</u>	Earned <u>Premium</u>
2012	2,400	1,000	400	5,000
2013	2,700	900		5,200
2014	2,100			5,400

a. (1.75 points)

Use the Neuhaus credibility weight to estimate the unpaid claim liability for accident year 2014 as a linear combination of R^{ind} and R^{coll} .

b. (0.5 point)

Assuming that $\text{Var}[U_i] = \text{Var}[U_i^{\text{BC}}]$ and using Hürlimann's method for optimal credibility and minimum variance, estimate the unpaid claim liability for accident year 2014.

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EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 1																	
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE: A1: Calculate unpaid claim estimates using credibility models.																
SAMPLE ANSWERS (BY PART, AS APPLICABLE)																	
Part a: 1.75 points																	
<i>Sample Answer 1</i>																	
$m_1 = 7,200 / 15,600 = 0.462$ $m_2 = 1,900 / (5,000 + 5,200) = 0.186$ $m_3 = 400 / 5,000 = 0.080$ $m_{\text{tot}} = 0.462 + 0.186 + 0.080 = 0.728$ $p_1 = m_1 / m_{\text{tot}} = 0.462 / 0.728 = 0.634$ $q_1 = 1 - p_1 = 0.366$ $R^{\text{ind}} = q_1 / p_1 \times C_{1,3}$ $= 0.366 / 0.634 \times 2,100 = 1,212$ $R^{\text{coll}} = q_1 \times U^{\text{BC}}$ $= q_1 \times m_{\text{tot}} \times V_1$ $= 0.366 \times 0.728 \times 5,400 = 1,438$ $Z^{\text{WN}} = m_1 = 0.462$ $R^{\text{C}} = Z^{\text{WN}} \times R^{\text{ind}} + (1 - Z^{\text{WN}}) \times R^{\text{coll}}$ $= 0.462 \times 1,212 + (1 - 0.462) \times 1,438 = 1,333$																	
<i>Sample Answer 2</i>																	
<table><tr><th>k</th><th>m_k</th><th>$p_k=m_k/\text{ELR}$</th><th>$Z^{\text{WN}} = p_1 * \text{ELR}$</th></tr><tr><td>1</td><td>$(24+27+21)/(50+52+54) = 0.462$</td><td>$.462/.728 = .634$</td><td>$0.634 * .728 = 0.462$</td></tr><tr><td>2</td><td>$(10+9)/(50+52)=.186$</td><td></td><td></td></tr><tr><td>3</td><td>$4/50=.08$</td><td></td><td></td></tr></table>	k	m_k	$p_k=m_k/\text{ELR}$	$Z^{\text{WN}} = p_1 * \text{ELR}$	1	$(24+27+21)/(50+52+54) = 0.462$	$.462/.728 = .634$	$0.634 * .728 = 0.462$	2	$(10+9)/(50+52)=.186$			3	$4/50=.08$			
k	m_k	$p_k=m_k/\text{ELR}$	$Z^{\text{WN}} = p_1 * \text{ELR}$														
1	$(24+27+21)/(50+52+54) = 0.462$	$.462/.728 = .634$	$0.634 * .728 = 0.462$														
2	$(10+9)/(50+52)=.186$																
3	$4/50=.08$																
$R^{\text{coll}} = 5,400 * (m_2 + m_3)=5,400 * (0.186 + 0.08) = 1,438$ $R^{\text{ind}} = 2,100 * (m_2 + m_3)/m_1=2,100 * (0.186 + 0.08)/0.462 = 1,212$ $R^{\text{C}} = Z^{\text{WN}} \times R^{\text{ind}} + (1 - Z^{\text{WN}}) \times R^{\text{coll}}$ $= 0.462 \times 1,212 + (1 - 0.462) \times 1,438 = 1,333$																	
Part b: 0.5 point																	
<i>Sample Answer 1</i>																	
$Z^* = p_1 / (p_1 + \sqrt{p_1}) = 0.634 / (0.634 + \sqrt{0.634}) = 0.443$ $R^{\text{C}} = Z^* \times R^{\text{ind}} + (1 - Z^*) \times R^{\text{coll}} = 0.443 \times 1,212 + (1 - 0.443) \times 1,438$ $= 1,338$																	
<i>Sample Answer 2</i>																	
$\text{LDF} = 1/p_k = 1/0.634 = 1.577$ $Z^* = (1/1.577) / ((1/1.577)+\sqrt{(1/1.577)})=0.443$ $R^{\text{C}} = Z^* \times R^{\text{ind}} + (1 - Z^*) \times R^{\text{coll}} = 0.443 \times 1,212 + (1 - 0.443) \times 1,438$ $= 1,338$																	

EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

EXAMINERS' REPORT

Candidates were expected to calculate the credibility-weighted reserves using both optimal credibility and Neuhaus credibility. The majority of candidates received full credit and demonstrated a clear understanding of the learning objectives. Of candidates who did not receive full credit, the errors were minor and included:

- Errors in the calculation (set up appears correct but calculation is wrong)
- Selecting the wrong credibility (Z was used for part a)
- Using the chain ladder approach to calculate R^{ind} & R^{coll}

These errors were more common in part a, as part b used the calculations from part a. Candidates understood the topic thoroughly.

Part a

The majority of candidates achieved full credit on this problem or made minimal errors. The candidate was expected to know how to calculate the estimated unpaid claim liability using the Neuhaus credibility. To accomplish this, they needed to perform the following:

- Calculate R^{ind} (need to correctly derive m_1 , ELR, p)
- Calculate R^{coll} (need to correctly derive m_1 , ELR, p)
- Calculate Neuhaus credibility
- Understand which estimate is the complement of credibility

Most candidates received full credit. The most common error was to incorrectly derive R^{ind} and R^{coll} using the chain ladder method. Other errors included:

- Incorrectly calculating R^{coll} as $U^{\text{BC}} - C_{1,3}$
- Weighting ultimates and then subtracting paid losses
- Applying the wrong credibility weights to R^{ind} and R^{coll}

Part b

The majority of candidates received full credit for this problem. The candidates were expected to recalculate the estimate of unpaid claim liability using optimal credibility instead of Neuhaus credibility estimate. They were expected to understand:

- Optimal Credibility
- Which estimate is the complement of credibility

Most candidates obtained full credit. One of the few errors seen was to use an incorrect credibility formula. Another was to write the correct formula but use a different value than p_1 in calculating credibility (such as m_1).